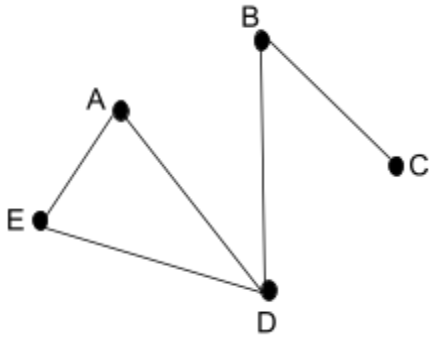
American Computer Science League

2021-2022 ● Contest 4: Short Problems ● Elementary Division

1. Graph Theory

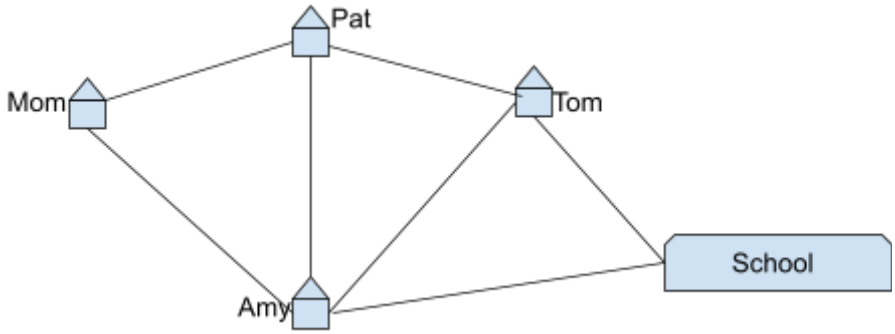
In the following graph, which edge could be removed to create an unconnected graph with at least one edge in each part?



A. AE
B. AD
C. ED
D. BD
E. BC

2. Graph Theory

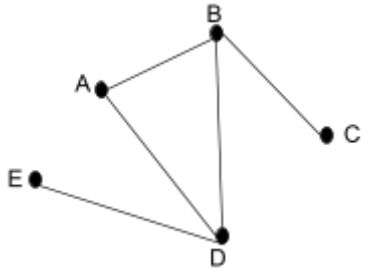
If your mom needed to pick up 3 friends (Amy, Pat, Tom) on your way to school using the following graph, how many different ways could she do it without going to any house more than once?



A. 2
B. 3
C. 4
D. 5
E. 6

3. Graph Theory

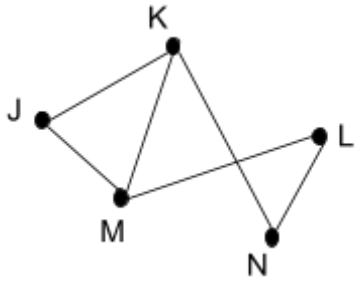
In a rumor network, a message is passed from one person to another through a third person. If this graph represents a rumor network among 5 good friends (Ann, Ben, Cal, Deb, Eva), which person(s) would be the middle person the most?



A. Ann
B. Ben
C. Deb
D. Ben, Deb
E. Ann, Ben, Deb

4. Graph Theory

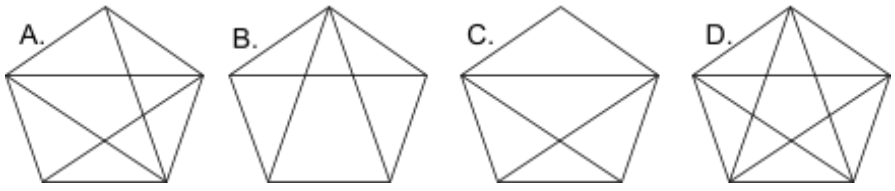
How many cycles are in the following graph?



A. 2
B. 3
C. 6
D. 8
E. 10

5. Graph Theory

Identify all of the following graphs that are traversable given that each graph only has 5 vertices on the outside pentagon.



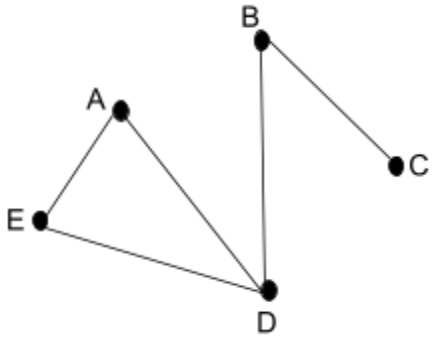
A. A, C, D
B. A, C
C. A, D
D. A, B, C
E. A, B, C, D

American Computer Science League

2021-2022 ● Contest 4: Short Problems ● Elementary Division

1. 图论

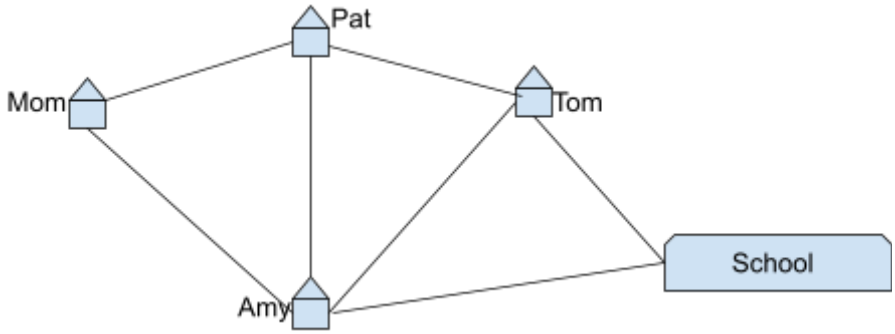
在下图中，删除哪条边**可以**得到**一个**非连通图，**且各**个部分**都**至少有一条边？



A. AE
B. AD
C. ED
D. BD
E. BC

2. 图论

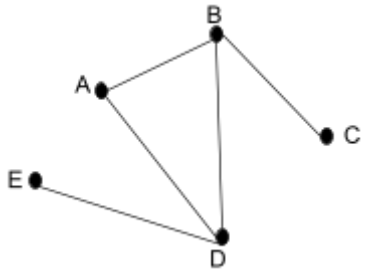
如果你的妈妈**需要**使用下面的**地图**，在你去学校的路上接三个朋友 (Amy, Pat, Tom) 去学校，如果每个房子**最多只能经过**一次，**请问妈妈有**多少种不同的方式接他们？



A. 2
B. 3
C. 4
D. 5
E. 6

3. 图论

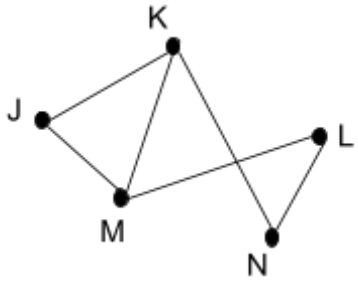
在**网络谣言**中，信息通过第三**方**从一个人传递给另一个人。如果这张图代表 5 个**好**朋友（Ann, Ben, Cal, Deb, Eva）之间的**谣言网络**，请问其中哪个人最可能是**位于**中间**的那个**人？



A. Ann
B. Ben
C. Deb
D. Ben, Deb
E. Ann, Ben, Deb

4. 图论

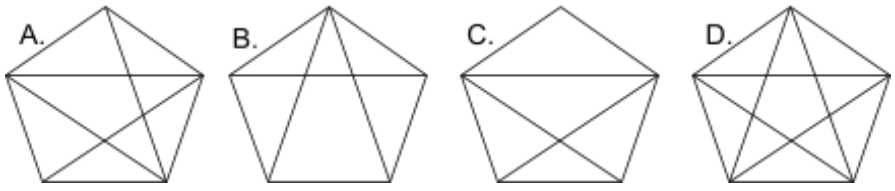
下图中有多少个环？



A. 2
B. 3
C. 6
D. 8
E. 10

5. 图论

假如每个图在外部的五边形上只有 5 个顶点，请**找出**下列所有可遍历的图。



A. A, C, D
B. A, C
C. A, D
D. A, B, C
E. A, B, C, D